

Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons

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by

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Approval

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Declaration

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Date: December 27, 2024

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Abstract

This thesis investigates methodologies for identifying and utilizing language-specific neurons in multilingual large language models (LLMs) to improve performance on multilingual downstream tasks, particularly for low-resource languages. The research examines how language-specific information is represented in various LLM architectures, including Llama3.1, Bloomz, and Aya models. Techniques such as activation probability entropy and LoRA-based fine-tuning methods were applied to identify language-specific neurons using datasets like Wikipedia and XNLI. Intervention strategies, such as replacing neuron activations with language-specific statistical activations, were explored to determine their effect on task performance. Experiments involving neuron-specific fine-tuning and neuron freezing were also conducted to study the relationship between architectural modifications and language representation.

The study further analyzes perplexity distributions and neuron activation maps across different languages and models, providing insights into how multilingual datasets influence the internal mechanisms of LLMs. Experimental setups were developed to assess the effectiveness of fine-tuning and intervention methods for improving natural language inference (NLI) performance in low-resource languages without requiring direct training data. By addressing existing gaps in language-specific neuron identification, this work contributes to a better understanding of how LLMs can be tailored to achieve performance improvement across diverse languages. The results provide a deeper understanding of the interaction between LLM architecture and multilingual capabilities, particularly for underrepresented languages.

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Abbreviations

AFT	A ccelerated F ailure T ime
ANOVA	A nalysis of V ariance
BLR	B inary L ogistic R egression

Symbols

μ	Location parameter
λ	Scale parameter
β_0	Null model parameter
$\hat{\beta}$	Fitted parameter

Chapter 1

Introduction

1.1 Introduction

The emergence of large language models (LLMs) has transformed natural language processing (NLP), enabling remarkable advancements across a wide range of linguistic tasks. Despite their success, challenges persist in ensuring comparable performance for low-resource languages, which often lack sufficient representation in pretraining datasets. This thesis addresses these challenges by investigating the representation of language-specific information within LLMs and exploring methods to identify and utilize language-specific neurons. By focusing on multilingual tasks such as natural language inference (NLI), this research seeks to better understand how language-specific information is encoded and how it can be effectively utilized to improve performance, particularly for underrepresented languages.

The research builds on recent advancements in neuron analysis and fine-tuning techniques, using models such as Llama3.1, Bloomz, and Aya to explore the relationship between architectural choices, pretraining datasets, and language-specific representations. By employing methods such as activation probability entropy and LoRA-based fine-tuning, the study systematically identifies language-specific neurons and evaluates their role in downstream tasks. Through detailed experiments on layer-specific fine-tuning, freezing, and intervention strategies, the thesis aims to provide a comprehensive understanding of the mechanisms underlying multilingual capabilities in LLMs.

1.2 Motivation

In the era of multilingual LLMs, achieving consistent performance across diverse languages remains a significant challenge. High-resource languages benefit from extensive pretraining data and resources, leading to superior performance on downstream tasks, while low-resource

languages often lag behind. Addressing this imbalance is essential for ensuring that the advancements in NLP benefit speakers of all languages, regardless of the resources available for their linguistic representation.

Understanding the internal mechanisms of LLMs, specifically how they represent language-specific information, offers a way to addressing these challenges. By identifying language-specific neurons and their roles in encoding linguistic knowledge, we can develop targeted interventions to improve performance for low-resource languages. This research is driven by the need to close the performance gap, advance the understanding of multilingual model architectures, and contribute to the development of fair and inclusive NLP systems.

1.3 Problem Statement

While LLMs exhibit strong multilingual capabilities, their performance across languages is often inconsistent, with low-resource languages showing suboptimal results compared to high-resource counterparts. This disparity can be attributed to the lack of sufficient representation for low-resource languages during pretraining and the challenge of effectively isolating and utilizing language-specific information within these models. Existing methods for identifying language-specific neurons often rely on arbitrary thresholds or lack generalizability across different architectures and datasets. Moreover, the impact of architectural modifications, such as layer-specific fine-tuning or neuron freezing, on language-specific representation and downstream performance remains underexplored.

This research aims to address these limitations by developing systematic approaches to identify and analyze language-specific neurons in LLMs. By focusing on techniques such as neuron activation probability and LoRA-based fine-tuning, the study seeks to bridge the gap in understanding how language-specific information is encoded. The ultimate goal is to improve task performance for low-resource languages without requiring extensive annotated data, thereby contributing to the application of LLMs in multilingual settings.

Chapter 2

Literature Review

2.1 Language-Specific Neurons in Multilingual Models

A significant body of work has focused on understanding and isolating language-specific neurons within large language models (LLMs). The paper *Language-Specific Neurons in Multilingual Language Models* ([ArXiv:2404.02431](#)) investigates techniques for identifying neurons that are specific to certain languages and evaluates their impact on tasks such as machine translation and multilingual understanding. Building on this, our research extends the idea by evaluating alternative methods such as replacing median activations with layer-specific median or 75th percentile activations and exploring whether language-specific neuron sets vary across datasets, tasks, and model layers.

The reproduction of results from the *Language-Specific Regions* paper ([ArXiv:2402.16438](#)) further validated the connection between perplexity changes and language specificity in LLMs. The study provided insights into how replacing or interpolating neuron activations affects task performance, particularly for low-resource languages. These findings underline the importance of understanding neuron activity patterns for improving multilingual tasks. For instance, experiments demonstrated that full replacement of language-specific neurons may degrade performance, while linearly interpolating activation values can balance task performance and neuron specificity.

2.2 Mechanistic Interpretability and Multilinguality

The broader context of neuron analysis in multilingual LLMs intersects with mechanistic interpretability, as explored in works such as the *SoLU Activation Function* ([SoLU paper](#)) and *Tracing the Roots of Facts in Multilingual Models* ([ArXiv:2403.05189](#)). The former highlights methods to reduce polysemanticity in neurons, which could aid in isolating neurons responsible for specific linguistic features. Meanwhile, the latter paper delves into probing techniques for

factual association in multilingual models, suggesting that neuron activity patterns can vary significantly depending on the dataset and task.

This thesis builds on these foundational ideas by experimenting with probing methods such as L1 and L2 norms and designing novel interventions in neuron activations. The literature also emphasizes how architectural changes, such as adding adapter layers or modifying attention heads, can enhance model performance on multilingual tasks. The exploration of neuron activity patterns for cross-lingual transfer, fact probing, and multilingual perplexity maps adds depth to the understanding of language representation in LLMs.

2.3 Applications to Low-Resource Languages

Addressing performance disparities for low-resource languages is a recurring theme across the literature. Papers such as *Locating and Editing Factual Associations in GPT* and *Discovering Salient Neurons in Deep NLP Models* highlight methodologies for identifying and modifying neurons to improve downstream task performance. This aligns with our research objectives, particularly in the context of improving natural language inference (NLI) and translation tasks for low-resource languages. Techniques such as activation replacement, layer freezing, and fine-tuning specific neurons have shown promise in achieving consistent task performance across diverse linguistic settings.

Chapter 3

Methodology

3.1 Definition of Neuron

In the context of multi layer perceptrons (MLPs) in large language models (LLMs), a *neuron* refers to a unit of computation within the MLP module. Specifically, consider the i^{th} layer of a Transformer architecture, where the input is a hidden state vector $\mathbf{H}^{i-1} \in \mathbb{R}^{T \times d}$, derived from the previous layer.

Hidden State Representation: The output of the multi-head self-attention (MHA) mechanism serves as input to the MLP and is given as:

$$\hat{\mathbf{H}}^i = \text{Attn}(\mathbf{H}^{i-1}\mathbf{W}_q^i, \mathbf{H}^{i-1}\mathbf{W}_k^i, \mathbf{H}^{i-1}\mathbf{W}_v^i) \cdot \mathbf{W}_o^i, \quad (3.1)$$

where $\mathbf{W}_q^i, \mathbf{W}_k^i, \mathbf{W}_v^i, \mathbf{W}_o^i \in \mathbb{R}^{d \times d}$ are the trainable parameters of the self-attention module, and $\text{Attn}(\cdot)$ denotes the scaled dot-product attention operation.

MLP Transformation: The MLP module takes $\hat{\mathbf{H}}^i$ as input and transforms it as follows:

$$\mathbf{h}^i = \text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \cdot \mathbf{W}_2^i, \quad (3.2)$$

where $\mathbf{W}_1^i \in \mathbb{R}^{d \times 4d}$ and $\mathbf{W}_2^i \in \mathbb{R}^{4d \times d}$ are learnable projection matrices, and $\text{act_fn}(\cdot)$ represents a non-linear activation function such as GELU or SiLU.

Neuron: Within the MLP, the j^{th} neuron of the i^{th} layer corresponds to the j^{th} row of the intermediate activation matrix:

$$\mathbf{h}_j^i = \text{act_fn}((\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j), \quad (3.3)$$

where $(\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j$ represents the pre-activation value of the j^{th} neuron.

Generalized Neuron Formulation: Recent architectures such as LLaMA utilize Gated Linear Units (GLU) to enhance MLP performance, modifying the MLP computation as:

$$\mathbf{h}^i = [\text{act_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (3.4)$$

where $\mathbf{W}_3^i \in \mathbb{R}^{d \times 4d}$ is an additional projection matrix, and $\otimes$ denotes element-wise multiplication. This formulation introduces gating to the MLP, allowing for more expressive transformations. Figure 3.1 shows the architecture of the MLP module.

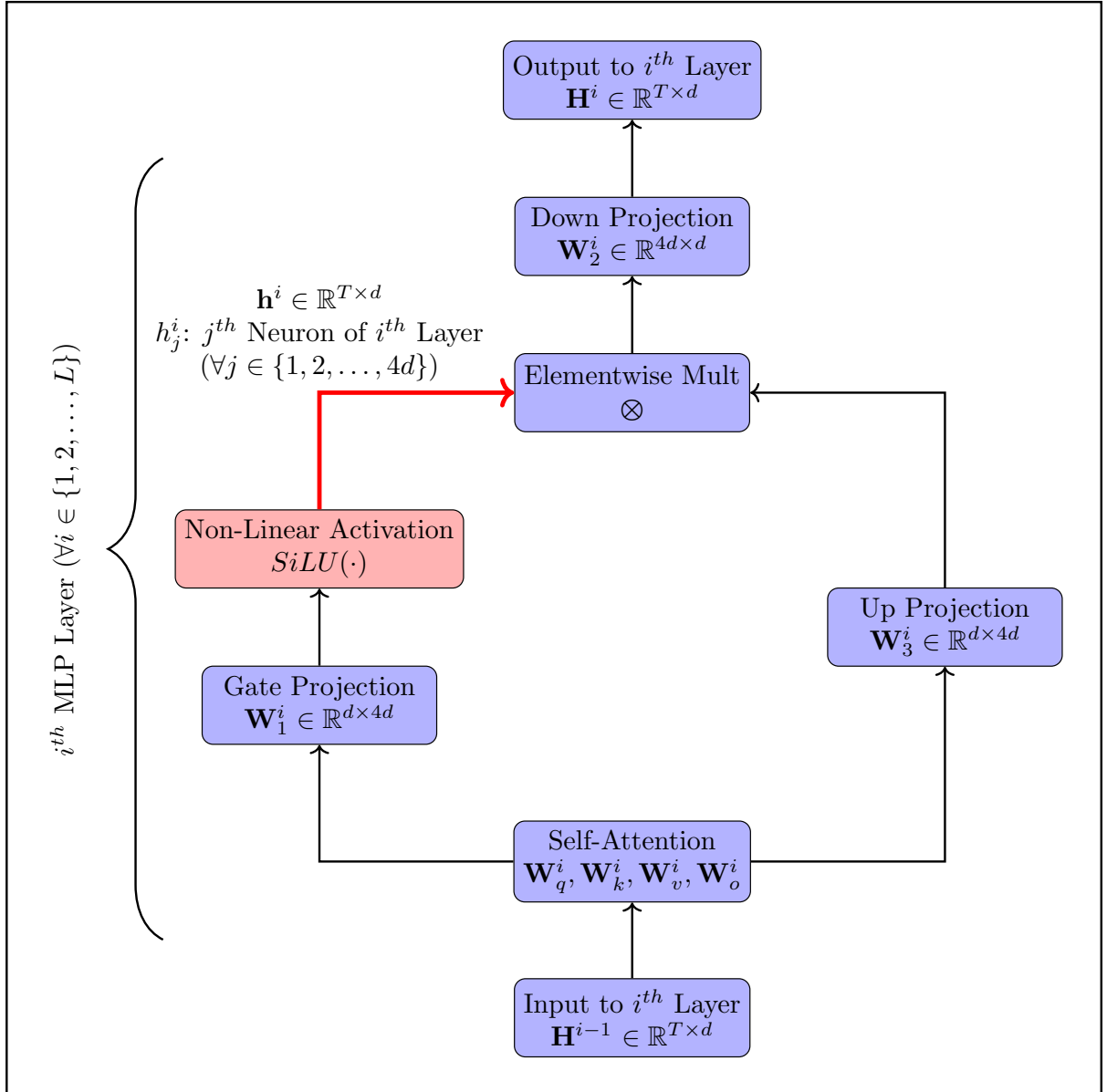


FIGURE 3.1: Architecture of MLP module in LLaMA model

3.2 Dataset for Neuron Identification

For the purpose of identifying language-specific neurons, we utilize a subset of the Wikipedia dataset spanning 15 languages:

$$\{\text{en, fr, es, vi, id, ja, zh, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

Each language's dataset is preprocessed and passed through pretrained large language models (LLMs). The activations are then collected from the MLP modules of the models.

Dataset Description: The Wikipedia dataset for each language contains a sequence of tokenized text data. Each tokenized sequence has a maximum length of $T_{\max} = 512$. For every sequence, we compute the activations from the MLP module, followed by averaging the activations at each neuron position across all sequences. This provides a language-specific profile for every neuron (i, j) in the model.

Activation Collection: Given a pretrained LLM, let the MLP activation for the i^{th} layer and the j^{th} neuron in a specific language be represented by $h_{i,j}^t \in \mathbb{R}$, where t indexes the token positions in a sequence of length T . The average activation of a neuron across all token positions for a given sequence is computed as:

$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.5)$$

where $h_{i,j}^t$ is the neuron activation corresponding to the t^{th} token.

Language-Specific Activations: For each language l in the dataset, we compute the average activation of each neuron (i, j) over all sequences S_l in the dataset:

$$H_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3.6)$$

where $H_{i,j}^l$ represents the average activation of the j^{th} neuron in the i^{th} layer for language l , and $|S_l|$ denotes the number of sequences for language l .

3.3 Statistics of Neuron Activation

The statistics of neuron activation are precomputed to analyze language-specific patterns in the MLP layers of the pretrained large language models (LLMs). These statistics include mean, standard deviation, and quantiles (e.g., 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles) of the neuron activations for each language in the dataset. The computations are performed across all layers and neurons within the MLP modules.

Activation Mean and Standard Deviation: The mean activation of a neuron (i, j) in the t^{th} token position is computed as:

$$\mu_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.7)$$

where $h_{i,j}^t$ is the activation of the (i, j) neuron for the t^{th} token, and T is the total number of tokens in a sequence. Similarly, the standard deviation of the activations is calculated as:

$$\sigma_{i,j} = \sqrt{\frac{1}{T} \sum_{t=1}^T (h_{i,j}^t - \mu_{i,j})^2}. \quad (3.8)$$

Quantiles of Activation: Quantile-based statistics provide insights into the distribution of activations for each neuron. The q -quantile activation value, denoted as P_q , is defined as:

$$P_q(h_{i,j}) = \inf \{x \in \mathbb{R} : F_{h_{i,j}}(x) \geq q\}, \quad (3.9)$$

where $F_{h_{i,j}}(x)$ is the cumulative distribution function (CDF) of the activations for neuron (i, j) . Commonly computed quantiles include:

$$P_{50}, P_{75}, P_{90}, P_{95}, P_{25}, P_{10}, P_5.$$

Language-Specific Aggregation: For a given language l and all sequences in the dataset, the mean and other statistics are aggregated across all token positions and batches. Let S_l denote the set of sequences for language l , and B represent the batch size. The aggregated mean activation for neuron (i, j) is computed as:

$$\bar{\mu}_{i,j}^l = \frac{1}{|S_l|} \sum_{b=1}^B \sum_{s \in S_l} \mu_{i,j}^{(s,b)}, \quad (3.10)$$

where $\mu_{i,j}^{(s,b)}$ is the mean activation of neuron (i, j) for the s^{th} sequence in the b^{th} batch.

3.4 Methods for Neuron Identification

3.4.1 LAPE

Language Activation Probability Entropy (LAPE) is a metric designed to identify neurons that exhibit language-specific behavior across multiple languages. This metric quantifies the entropy of normalized activation probabilities for each neuron across the given set of languages.

Step 1: Compute Activation Probabilities. For a pretrained large language model with L layers and $4d$ neurons per layer, the activation of a neuron (i, j) in response to a sequence $X^l \in \mathbb{R}^{N \times T}$ (where N is the number of sequences, and T is the number of tokens per sequence)

for a language l is denoted as $h_{i,j}^l$. The unnormalized activation probability for neuron (i, j) is given by:

$$p_{i,j}^l = \mathbb{E} [\mathbb{I} (h_{i,j}^l > 0)] , \quad (3.11)$$

where $\mathbb{I}$ is the indicator function that returns 1 if the activation $h_{i,j}^l$ is greater than zero, and 0 otherwise. The expectation is computed over all tokens in all sequences for language l .

Step 2: Normalize Activation Probabilities. The normalized activation probability is computed across k languages to ensure a valid probability distribution:

$$P_{i,j}^l = \frac{p_{i,j}^l}{\sum_{l=1}^k p_{i,j}^l}, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.12)$$

Step 3: Compute Language Activation Probability Entropy. The entropy of the normalized activation probabilities for each neuron (i, j) is computed as:

$$\text{LAPE}(i, j) = - \sum_{l=1}^k P_{i,j}^l \log P_{i,j}^l, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.13)$$

This measures the uncertainty in the activation distribution of a neuron across the k languages. Figure 3.2 shows the procedure of computing LAPE for a neuron.

Step 4: Apply Threshold and Select Neurons. To identify neurons with significant language-specific behavior, a threshold is defined based on the 95th percentile of the unnormalized activation probabilities:

$$\text{Threshold} = \text{Quantile}_{0.95} (p_{i,j}^l \mid \forall i, j, l). \quad (3.14)$$

A neuron (i, j) is considered for language-specific analysis only if:

$$\exists l \in \{1, \dots, k\} \text{ such that } p_{i,j}^l > \text{Threshold}. \quad (3.15)$$

Step 5: Select Top Neurons Based on LAPE. The top m neurons with the lowest LAPE values are selected for language-specific analysis, where m is defined as:

$$m = \frac{0.01 \times 4d \times L}{100}. \quad (3.16)$$

This ensures that only the most language-specific neurons are retained for further interpretation.

Step 6: Identify Language-Specific Neuron Sets. For each neuron (i, j) in the selected set, the language-specific set is defined as:

$$\text{Lang_Spec_Set}(i, j) = \{l \mid P_{i,j}^l > \text{Threshold}\}. \quad (3.17)$$

A neuron is considered specific to a subset of languages if:

$$\text{Lang_Spec_Set}(i, j) \subseteq \{1, 2, \dots, k\}. \quad (3.18)$$

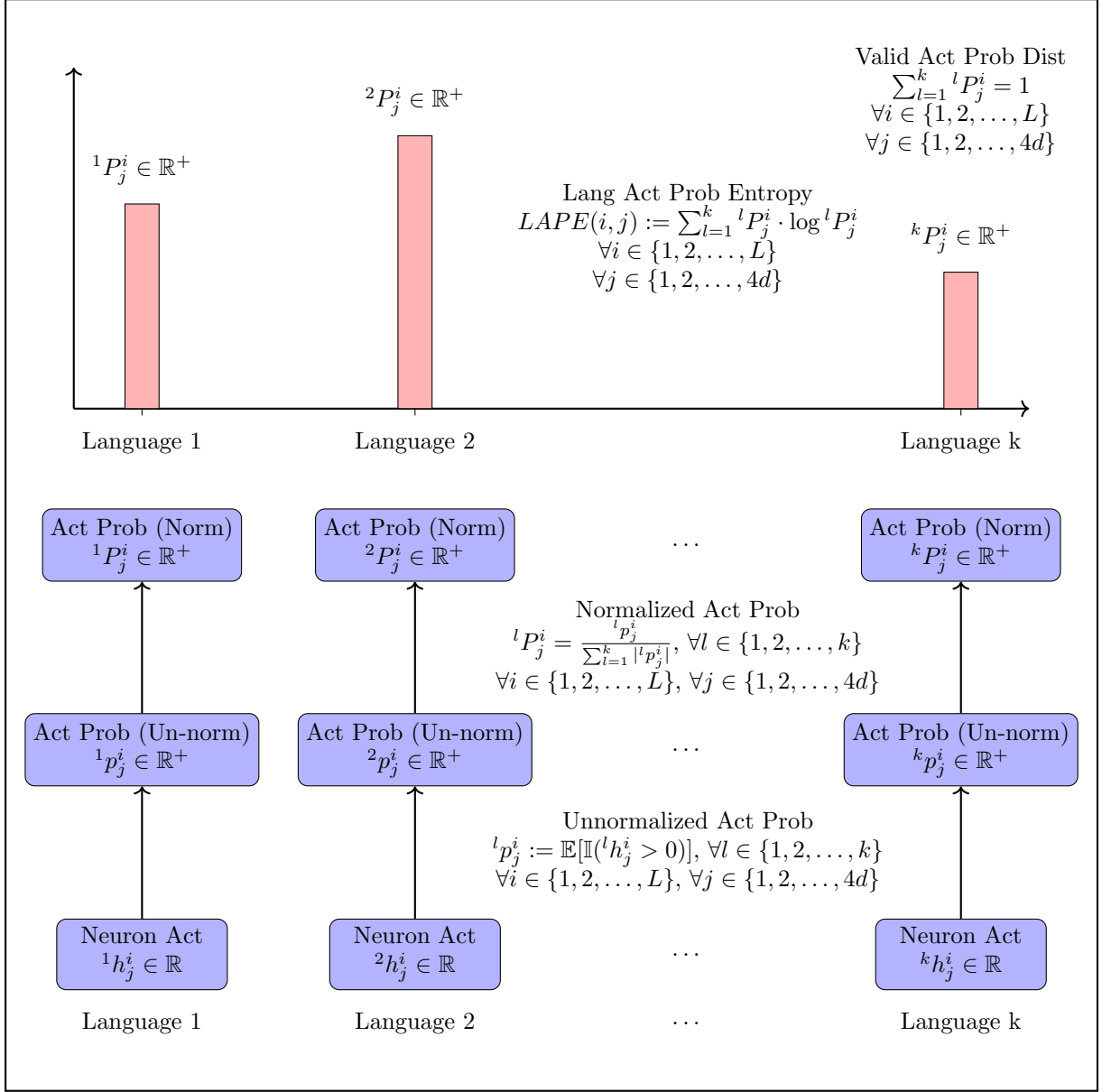


FIGURE 3.2: Procedure of computing LAPE for a neuron.

3.4.2 Activation Based Methods

Absolute Activation: The absolute activation metric measures the mean absolute activation of each neuron (i, j) in the i^{th} layer across all tokens in a sequence for a language l . This is computed as:

$$r_{i,j}^l = \mathbb{E} [|h_{i,j}^l|], \quad (3.19)$$

where $h_{i,j}^l$ is the activation of neuron (i, j) for language l , and the expectation is taken over all tokens and sequences. This metric identifies neurons with consistently high absolute activations, which are indicative of their overall importance in representing language features.

Activation Probability Zero: The activation probability zero metric measures the fraction of times a neuron (i, j) has a positive activation for a language l . It is defined as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I} (h_{i,j}^l > 0)], \quad (3.20)$$

where $\mathbb{I}$ is the indicator function that returns 1 if $h_{i,j}^l > 0$, and 0 otherwise. This metric captures the frequency of a neuron's activation, providing insights into its contribution to language-specific representations.

Activation Probability Mean: The activation probability mean measures the probability of a neuron's activation exceeding the mean activation value for its layer, $\mu_{i,j}^l$, for a language l . It is computed as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I} (h_{i,j}^l > \mu_{i,j}^l)], \quad (3.21)$$

where $\mu_{i,j}^l$ represents the average activation of neuron (i, j) in the i^{th} layer for language l . This metric identifies neurons with activations that are consistently above the average, highlighting neurons that are highly responsive to language-specific features.

Activation Probability 90p: The activation probability 90p measures the probability of a neuron's activation exceeding the 90th percentile activation value for its layer, $\text{P90}_{i,j}^l$, for a language l . This is given by:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I} (h_{i,j}^l > \text{P90}_{i,j}^l)], \quad (3.22)$$

where $\text{P90}_{i,j}^l$ represents the 90th percentile of the activations of neuron (i, j) in the i^{th} layer for language l . This metric identifies neurons that exhibit rare but significant activations, which may correspond to specialized processing for certain linguistic features.

Language-specific neurons are identified based on activation values computed across different languages. For each language, activations are collected, and neurons that exhibit the highest relevance are selected using the following process.

Step 1: Compute Activation Relevance. For each language $l \in \{1, 2, \dots, k\}$, compute the activation relevance values for all neurons (i, j) in the pretrained model. Let the relevance for neuron (i, j) in the i^{th} layer for language l be denoted as $r_{i,j}^l$. This relevance is calculated using one of the activation probability methods, such as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I} (h_{i,j}^l > 0)], \quad (3.23)$$

or other methods such as absolute activation or percentile-based thresholds, depending on the desired analysis.

Step 2: Aggregate Relevance Across Layers and Neurons. For each language l , the computed relevance values $r_{i,j}^l$ are aggregated across all layers and neurons. The total number of layers is denoted by L , and the dimensionality of the neurons in each layer is $4d$. The relevance

tensor for language l is represented as:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\}, j \in \{1, \dots, 4d\}\}. \quad (3.24)$$

Step 3: Select Top Neurons. For each language l , the top m neurons with the highest relevance values are selected. The value of m is determined as a small fraction of the total number of neurons in the model:

$$m = \frac{0.125 \times L \times 4d}{100}. \quad (3.25)$$

The top m neurons are identified by sorting the relevance values in R^l and selecting the indices of the m largest values:

$$\text{Top}_m(R^l) = \text{argsort_descending}_m \{r_{i,j}^l \mid \forall i, j\}. \quad (3.26)$$

Step 4: Map Neurons to Layers and Positions. The indices of the top m neurons are mapped back to their corresponding layers and positions. For each language l , the selected neurons are represented as:

$$N^l = \{(i, j) \mid (i, j) \in \text{Top}_m(R^l)\}, \quad (3.27)$$

where N^l is the set of top m neurons for language l .

This process ensures that neurons with the highest language-specific relevance are identified for each language. These neurons can then be analyzed further to understand their behavior and contribution to language modeling tasks.

3.5 Neuron Fine-Tuning using LoRA for a Task

LoRA (Low-Rank Adaptation): LoRA introduces a mechanism to fine-tune a subset of parameters in neural networks, focusing on reducing computational overhead. For our application, we utilize LoRA to specifically fine-tune targeted neurons in the MLP layer while freezing others. This method ensures that the pretrained parameters W_{pt} remain untouched, and ΔW represents the modifications applied to the network during fine-tuning.

Fine-Tuning with LoRA AB Decomposition. The matrix ΔW is expressed as:

$$\Delta W = A \cdot B, \quad (3.28)$$

where $A \in \mathbb{R}^{d_{in} \times r}$ and $B \in \mathbb{R}^{r \times d_{out}}$ are low-rank matrices. The rank r is typically much smaller than d_{in} or d_{out} , reducing the number of trainable parameters. Masks are applied to A and B to control the fine-tuning of specific neurons:

$$A = \text{masked_A} \cdot A_t, \quad B = \text{masked_B} \cdot B_t, \quad (3.29)$$

where masked_A and masked_B are binary masks, and A_t and B_t are trainable parameters.

Description of Masked Matrices:

The binary masks masked_A and masked_B are defined to control the fine-tuning process by selectively freezing specific neurons.

$$\text{masked_A} \in \mathbb{R}^{d_{in} \times r}, \quad \text{masked_B} \in \mathbb{R}^{r \times d_{out}},$$

where d_{in} and d_{out} are the input and output dimensions of the layer, and r is the rank of the LoRA decomposition.

Masked Matrix masked_A : masked_A is defined as:

$$[\text{masked_A}]_{ij} = 1, \quad \forall i \in \{1, \dots, d_{in}\}, \forall j \in \{1, \dots, r\}. \quad (3.30)$$

This indicates that no masking is applied to the rows of A , and all elements of masked_A are set to 1.

Masked Matrix masked_B : For masked_B , specific columns corresponding to frozen neurons are set to 0. For a neuron index $j_f \in \text{Frozen Neurons}$, the mask is defined as:

$$[\text{masked_B}]_{ij} = \begin{cases} 0, & \text{if } j = j_f, j_f \in \text{Frozen Neurons}, \\ 1, & \text{otherwise.} \end{cases} \quad (3.31)$$

Final Masking Condition: The effective masked matrices for A and B are:

$$A = \text{masked_A} \cdot A_t, \quad B = \text{masked_B} \cdot B_t, \quad (3.32)$$

where $A_t \in \mathbb{R}^{d_{in} \times r}$ and $B_t \in \mathbb{R}^{r \times d_{out}}$ are the trainable LoRA matrices.

This masking ensures that updates to the frozen neurons are completely suppressed, while other neurons remain trainable during fine-tuning.

Frozen Neurons. To freeze specific neurons, columns of B corresponding to the frozen neurons are masked with zeros. For neuron indices j in the set of frozen neurons, we ensure:

$$\Delta W_{i,j} = 0, \quad \forall i. \quad (3.33)$$

This prevents any updates to these neurons during fine-tuning.

Final Output. The fine-tuned output of the MLP layer is given by:

$$y = x \cdot W + x \cdot \Delta W, \quad (3.34)$$

where W is the original weight matrix, and ΔW introduces targeted updates through LoRA.

Targeted Language Neurons. Language-specific neurons are identified and fine-tuned using LoRA. This targeted approach ensures that only the relevant neurons for a particular language

are adapted while the others remain unchanged. For the frozen neurons, ΔW remains zero:

$$\Delta W_{i,j} = 0, \quad \text{if } j \in \text{Frozen Neurons.} \quad (3.35)$$

Additional Fine-Tuning. Besides fine-tuning the MLP layer, the attention mechanisms (q, k, v, o projections) and the task-specific classifier head are always fine-tuned using LoRA. This ensures the overall task-specific performance is preserved while leveraging the language neurons for targeted adaptation. Figure 3.3 shows the procedure of finetuning target neurons using LoRA.

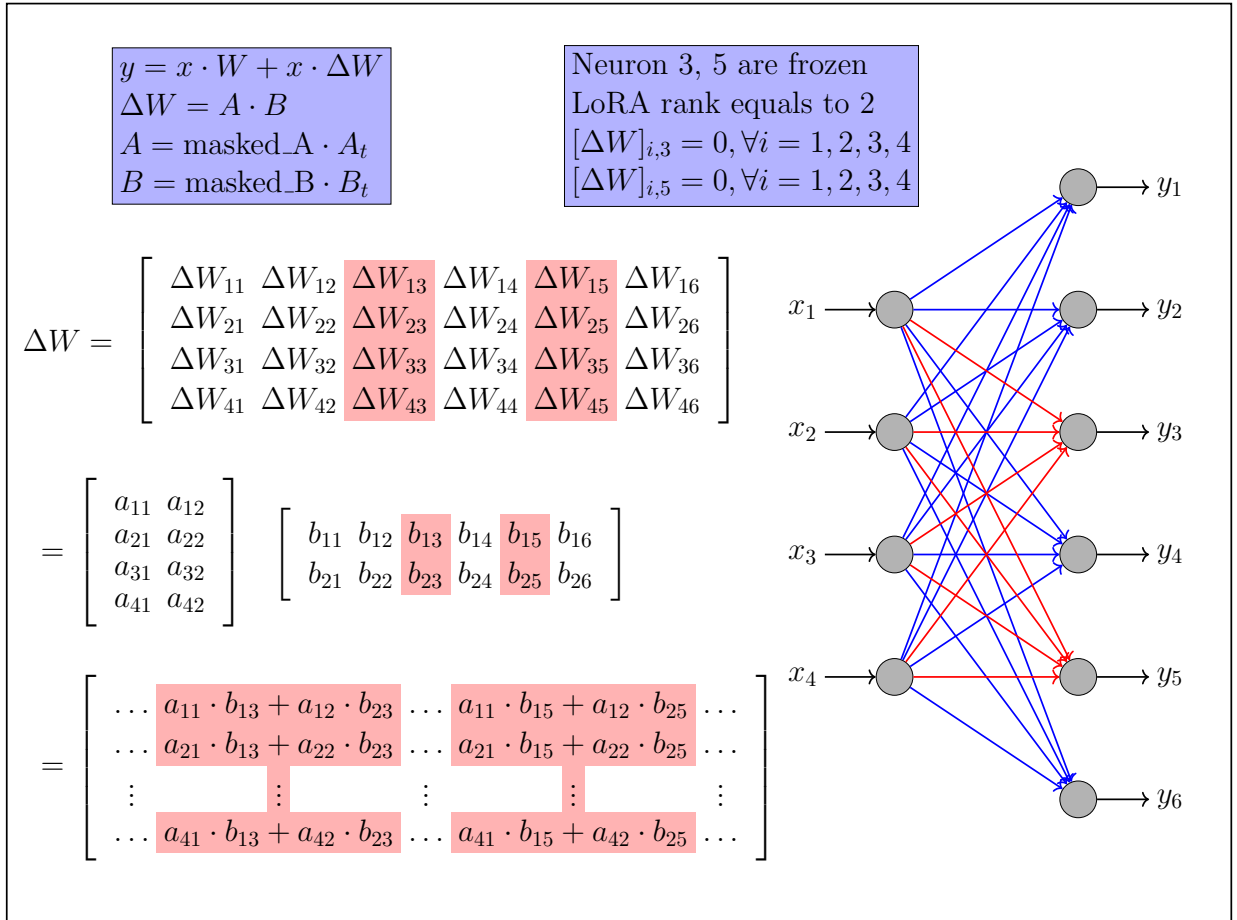


FIGURE 3.3: LoRA fine tuning method for target neurons.

Chapter 4

Experiments

4.1 Task and Dataset: XNLI

The Cross-lingual Natural Language Inference (XNLI) task is a benchmark designed to evaluate the capability of large language models (LLMs) to perform reasoning across multiple languages. The goal is to classify the relationship between a premise and a hypothesis into one of three categories: *entailment*, *neutral*, or *contradiction*. This task tests the ability of the model to generalize semantic relationships beyond the English language, making it a challenging benchmark for multilingual models.

Dataset Overview: The XNLI dataset contains annotated sentence pairs across 15 languages. In our experiments, we focus on English (*en*) as the source language and Vietnamese (*vi*), Hindi (*hi*) and Urdu (*ur*) as target languages. The dataset for each language can be represented as:

$$\mathcal{D}_l = \{(x_p^i, x_h^i, y^i) \mid x_p^i, x_h^i \in \mathcal{X}_l, y^i \in \{0, 1, 2\}\}, \quad (4.1)$$

where x_p^i and x_h^i denote the premise and hypothesis sentences in language l , and y^i is the corresponding label indicating the relationship.

Tokenization and Preprocessing: Each sentence pair (x_p^i, x_h^i) is concatenated with a special token to form the input:

$$x^i = x_p^i + [\text{EOS}] + x_h^i, \quad (4.2)$$

where $[\text{EOS}]$ represents the end-of-sequence token. The concatenated input is tokenized into a sequence of tokens $\{t_1, t_2, \dots, t_T\}$ with a maximum sequence length $T = 512$.

Train and Test Splits: The dataset is divided into training and testing subsets for each language:

$$\mathcal{D}_l = \mathcal{D}_l^{\text{train}} \cup \mathcal{D}_l^{\text{test}}, \quad \mathcal{D}_l^{\text{train}} \cap \mathcal{D}_l^{\text{test}} = \emptyset. \quad (4.3)$$

For the experiments, a fraction $\mathcal{F} \subset \mathcal{D}_l^{\text{train}}$ is used to train the model. The size of $\mathcal{F}$ is determined as:

$$|\mathcal{F}| = \text{frac} \times |\mathcal{D}_l^{\text{train}}|, \quad (4.4)$$

where frac is the fraction of the training dataset used.

Multilingual Challenges: In the XNLI task, transferring knowledge from English to other languages like Vietnamese and Hindi requires the model to overcome both semantic and structural differences in language. This transfer is tested by evaluating the zero-shot performance on the target languages without directly training on their data. The objective of this task is to fine-tune LLaMA3, a large language model, on the XNLI dataset to improve cross-lingual inference capabilities. The experiments are designed to test the effectiveness of various fine-tuning setups and analyze the role of language-specific neurons in improving performance across multiple languages.

4.2 Baseline Experiments

The baseline experiments establish a foundation for evaluating the performance of different fine-tuning setups on the XNLI task. We include two baselines: zero-shot evaluation of a model fine-tuned on English (*en*) XNLI data and direct fine-tuning on the target language (*vi*) XNLI data. These baselines provide a comparative framework to assess the impact of fine-tuning specific language neurons using LoRA.

Zero-Shot Baseline: The first baseline involves fine-tuning the pretrained model on the English XNLI dataset. Let $\mathcal{M}_{\text{en}}$ denote the model fine-tuned on English data. The zero-shot evaluation involves testing $\mathcal{M}_{\text{en}}$ on Vietnamese XNLI datasets without further training. The evaluation metric, accuracy, is defined as:

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(\hat{y}^i = y^i)}{N}, \quad (4.5)$$

where N is the total number of test examples, y^i is the ground truth label, $\hat{y}^i$ is the predicted label, and $\mathbb{I}(\cdot)$ is the indicator function.

Direct Fine-Tuning Baseline: The second baseline involves directly fine-tuning the pretrained model on the Vietnamese XNLI dataset. Let $\mathcal{M}_{\text{vi}}$ denote the model fine-tuned on Vietnamese data. This baseline establishes an upper bound for performance on the Vietnamese XNLI task, as the model is explicitly trained on the target language data. We again use the accuracy as evaluation metric.

4.3 Experiment Goals

The primary objective of this work is to improve the zero-shot performance of the model on the target language, such as Vietnamese (*vi*), without utilizing any target language training data. This constraint is critical to ensure that the model remains agnostic to the specific characteristics of the target language during training, thereby simulating a real-world cross-lingual transfer learning scenario.

Baseline Comparison: In this setup, we consider two baselines:

- **Baseline 1: Zero-Shot Performance on *vi* using English Training Data.** This baseline involves fine-tuning the pretrained model on the English (*en*) XNLI dataset ($\mathcal{M}_{\text{en}}$) and evaluating its zero-shot performance on the Vietnamese (*vi*) test set. This represents the starting point of cross-lingual transfer without any modifications to the model architecture or targeted interventions.
- **Baseline 2: Performance on *vi* using Direct Fine-Tuning on *vi* Data.** This baseline involves directly fine-tuning the pretrained model on the Vietnamese (*vi*) XNLI dataset ($\mathcal{M}_{\text{vi}}$). It provides an upper bound on performance since it leverages target language training data.

Objective Constraints: While the ultimate goal is to improve the zero-shot performance on the *vi* test set, it is imperative to operate under the following constraints:

1. No target language (*vi*) training data can be used during the fine-tuning process.
2. Only the English (*en*) XNLI training data is available for model training and intervention.
3. The interventions must be limited to model surgery techniques, such as fine-tuning specific neurons or modifying activation patterns.

Target Metric: The improvement in zero-shot performance is quantified as:

$$\Delta_{\text{ZS}} = \text{Accuracy}(\mathcal{M}_{\text{experiment}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}), \quad (4.6)$$

where $\mathcal{M}_{\text{experiment}}$ refers to the model modified through various intervention strategies, and $\mathcal{M}_{\text{en}}$ refers to the baseline model fine-tuned on *en*.

Goal: The overarching goal is to maximize Δ_{ZS} , thereby surpassing the performance of Baseline 1 ($\mathcal{M}_{\text{en}}$) through innovative techniques such as:

- Fine-tuning language-specific neurons identified for the target language (*vi*) using only the *en* training data.
- Utilizing statistical activation patterns (*e.g.*, *mean*, *quantile-based thresholds*) to intervene in neuron activations during inference.
- Applying LoRA (Low-Rank Adaptation) to selectively fine-tune specific components such as *qkv* projections and classification heads while freezing other parts of the model.

Limitations: By design, the performance of any intervention cannot exceed Baseline 2 ($\mathcal{M}_{\text{vi}}$), as that baseline uses direct training data from the target language. However, our approach aims to close the gap between Baseline 1 ($\mathcal{M}_{\text{en}}$) and Baseline 2 ($\mathcal{M}_{\text{vi}}$) as much as possible through zero-shot strategies.

4.4 Targeted Neuron Fine-tuning Experiment

This section outlines the experimental setups designed to improve the zero-shot performance of the model on the Vietnamese (*vi*) XNLI dataset. Four configurations are explored, each using different strategies to fine-tune the model, while adhering to the constraint of using only English (*en*) training data during fine-tuning. The experiments aim to utilize targeted interventions to improve performance by selectively tuning neurons and applying LoRA-based adaptations.

Setup A: Standard Fine-Tuning with LoRA. In this setup, the pretrained model is fine-tuned on the English (*en*) XNLI dataset using LoRA. The fine-tuning is applied to the query (q), key (k), value (v), and output (o) projection matrices ($W_q^i, W_k^i, W_v^i, W_o^i$) and the classification head. The resulting model is evaluated on the Vietnamese (*vi*) dataset in a zero-shot setting. Additionally, the *vi*-specific activations are substituted by statistical activation patterns such as mean (μ), 90th percentile, and others, defined as:

$$\hat{h}_j^{i,vi} = \text{stat}_j^{i,vi}, \quad \forall \text{stat} \in \{\text{mean}, 90p, 95p, 75p, 5p, 10p, 25p, 0\}, \quad (4.7)$$

where $\hat{h}_j^{i,vi}$ represents the substituted activation for *vi* language for neuron (i, j) in the MLP block.

Setup B: Fine-Tuning with LoRA and English Neurons. This setup builds on Setup A by incorporating fine-tuning of English-specific neurons identified in the MLP block. Let $\mathcal{N}_{\text{en}}$ denote the set of English-specific neurons. During fine-tuning, the LoRA modifications (ΔW) are applied to both the query-key-value-output projections and the English neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{en}}. \quad (4.8)$$

This targeted approach allows the model to specialize in English while preserving the generalization capacity of other neurons.

Setup C: Fine-Tuning with LoRA and Target Language Neurons. In this configuration, the focus shifts to fine-tuning target language such as Vietnamese-specific neurons ($\mathcal{N}_{\text{vi}}$). During fine-tuning, the LoRA modifications (ΔW) are applied to both the query-key-value-output projections and the Vietnamese neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{vi}}. \quad (4.9)$$

This ensures that only the Vietnamese-specific neurons are fine-tuned while others remain untouched.

Setup D: Fine-Tuning with LoRA and Both English & Target Language Neurons. This setup combines the interventions from Setup B and Setup C, targeting both English and Vietnamese neurons. Let $\mathcal{N}_{\text{en,vi}} = \mathcal{N}_{\text{en}} \cup \mathcal{N}_{\text{vi}}$ represent the union of English and Vietnamese neurons. During fine-tuning, the LoRA modifications (ΔW) are applied to both the query-key-value-output projections and both the English & Vietnamese neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{en,vi}}. \quad (4.10)$$

This strategy allows the model to simultaneously adapt to English and Vietnamese language-specific representations.

Comparative Analysis: After fine-tuning, the zero-shot performance on Vietnamese (*vi*) XNLI data is measured using the accuracy metric. The performance improvements achieved through the setups are quantified by:

$$\Delta_{\text{zs}} = \text{Accuracy}(\mathcal{M}_{\text{setup}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}), \quad (4.11)$$

where $\mathcal{M}_{\text{setup}}$ refers to the model fine-tuned using one of the experimental setups. These improvements are compared against the baselines to evaluate the efficacy of the targeted neuron fine-tuning strategies.

4.5 LAPE Experiments

4.5.1 Language Sets

4.5.2 Language Neuron Distribution

4.5.3 Perplexity Change

4.6 Activation Probability 90p Experiments

Chapter 5

Analysis and Results

5.1 Introduction

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5.2 Descriptive Statistics

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5.3 Model Development

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5.5 Summary

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Chapter 6

Conclusions and Practical Implications

6.1 Summary

This chapter outlines the findings and conclusions of the study, as well as the practical implications that can be drawn from the results.

6.2 Study Findings

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6.3 Research Contribution

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6.4 Practical Implications

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6.5 Study Limitations and Scope for Future Work

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Appendix A

Appendix A Title

Appendix B

Appendix B Title

References

List of Publications

Articles Published in International/National Journals

1. Article item 1
2. Article item 2

Presentations and Proceedings in International/National Conferences

1. Conference item 1
2. Conference item 2